

An AI-Driven Automated Interview Platform with Avatar-Based Candidate Assessment

Design, Implementation, and Evaluation

Shlok Abhishek

Independent Researcher

India

Email: shlokabhishek@example.com

Abstract—The rapid growth of remote hiring has created demand for interview systems that scale without sacrificing consistency or candidate experience. This paper presents *AI Interview Platform*, a web-based system that lets recruiting teams run automated, asynchronous video interviews through a personalized AI avatar. The platform combines a React single-page application with a serverless backend, and uses a large language model (LLM) for question generation, response evaluation, and candidate ranking, with a deterministic rule-based engine as a fallback whenever the LLM is unavailable. A distinguishing feature is the ability to synthesize an interviewer avatar from an HR professional’s photographs and voice, paired with text-to-speech (TTS) and speech-to-text (STT) for a conversational interview flow. The system also includes a lightweight, privacy-preserving computer-vision heuristic that flags likely phone use during recording. We describe the architecture, a five-dimension scoring model (relevance, accuracy, confidence, keyword coverage, and skill signal), and the graceful-degradation strategy that keeps the platform operational under API outages or rate limits. We report a functional and qualitative evaluation of the implemented system and, because automated hiring tools now fall under active legal and regulatory scrutiny (e.g., NYC Local Law 144, the Illinois AI Video Interview Act, and the EU AI Act’s high-risk classification of employment AI), we discuss the fairness, transparency, and auditability requirements the platform would need to satisfy before production deployment. We position this as a systems and design contribution rather than a validated hiring instrument, and we outline the empirical study needed to establish predictive validity and demographic fairness prior to real-world use.

Index Terms—Artificial Intelligence, Large Language Models, Automated Interviewing, AI Avatar, Speech Processing, Candidate Assessment, Algorithmic Fairness, Human-Computer Interaction

I. INTRODUCTION

Traditional hiring processes depend heavily on human judgment, which creates well-documented challenges around scalability, consistency, and objectivity. As organizations grow, HR teams are often overwhelmed by application volume, leading to long hiring cycles, scheduling bottlenecks, and interviewer-to-interviewer variability in scoring [1], [2]. Structured interviews reduce some of this variability but remain labor-intensive and difficult to standardize across many interviewers and requisitions.

Advances in natural language processing (NLP) and large language models (LLMs) have made it feasible to automate

large parts of the recruitment pipeline: generating context-aware questions, evaluating open-ended responses semantically, and producing evidence-based scores. However, most existing automated screening tools present candidates with static forms or text-based chatbots, which strips away the interpersonal cues that matter for assessing communication and interpersonal skills, and can leave candidates feeling that they are being evaluated by an opaque system rather than engaged in a conversation.

This paper describes *AI Interview Platform*, a web system that aims to preserve more of the human interview experience while automating the surrounding workflow. An AI avatar, trained from an interviewer’s photographs and voice samples, delivers questions via TTS and video, and candidates respond through the browser’s microphone and camera. The system automates the interview lifecycle end to end: session creation, question generation, candidate registration, response capture, semantic scoring, and ranking.

At the same time, automated video-interview tools have drawn substantial legal and civil-rights scrutiny in recent years. Vendors such as HireVue faced a 2019 FTC complaint from the Electronic Privacy Information Center (EPIC) alleging that its facial-analysis scoring was unprovable and biased, and the company removed facial analysis from its product in January 2021 [7]. More recently, the ACLU of Colorado filed an EEOC and state civil-rights complaint on behalf of a deaf, Indigenous former Intuit employee, alleging that an AI-scored video interview penalized her communication style and that a request for human-generated captioning as an accommodation was denied [8], [9]. Separately, New York City’s Local Law 144 now requires employers using automated employment decision tools to commission independent annual bias audits, publish impact-ratio results, and give candidates advance notice [10], Illinois has required disclosure and consent for AI-analyzed video interviews since 2020 [11], and the EU AI Act classifies employment-related AI systems as high-risk, subjecting them to transparency, accuracy, and human-oversight obligations [11]. This regulatory context directly shapes what a system like ours must support — disclosure, auditability, and human oversight — and we return to it in Section VII.

The primary contributions of this work are as follows.

- We design and implement a full-stack, serverless-ready

AI interview platform with separate interviewer and candidate portals.

- We propose a personalized AI avatar pipeline that combines static imagery, optional video, and TTS to deliver a more conversational interview experience than form- or chatbot-based screening.
- We develop a dual-mode question-generation engine: an LLM-driven mode that produces role- and resume-tailored questions, and a deterministic rule-based mode that keeps the system functional without external AI connectivity.
- We present a five-dimension scoring methodology (relevance, accuracy, confidence, keyword coverage, and skill presence) with a matching rule-based fallback analyzer, so that scoring degrades gracefully rather than failing outright.
- We implement a client-side, privacy-preserving heuristic for flagging likely unauthorized phone use during recording, without transmitting camera frames to a server.
- We discuss, in light of current litigation and regulation (Local Law 144, the Illinois AI Video Interview Act, the EU AI Act), the auditing, disclosure, and accommodation mechanisms that would be required before such a system could be responsibly deployed in production hiring.

The remainder of this paper is organized as follows. Section II reviews related work on automated interviewing and on the regulatory response to AI hiring tools. Section III describes the system architecture. Section IV details the core methodology. Section V presents implementation specifics. Section VI reports our evaluation and its limitations. Section VII discusses fairness and compliance requirements. Section VIII concludes and outlines future work.

II. RELATED WORK

A. Automated Interview Scoring

Early automated interview systems relied on structured question banks and keyword matching [3]. These approaches are inexpensive to run but cannot capture semantic nuance or reward well-phrased answers that avoid the exact expected keywords. Transformer-based language models, from BERT to instruction-tuned GPT-family models, have since been applied to semantic similarity scoring and open-ended answer evaluation [4], [5], enabling systems to move beyond literal keyword overlap. Our platform builds on this line of work by using an instruction-tuned LLM for both question generation and holistic, multi-dimensional scoring, while retaining a deterministic scorer as a fallback for reliability.

B. Avatars and Conversational Agents

Text-to-speech and computer-vision techniques have enabled digital humans and virtual agents in customer service, training, and tutoring contexts [6]. The application of a *personalized* avatar, trained on a specific HR professional’s likeness and voice, to the interview domain is comparatively underexplored. Our system contributes to this space by pairing

avatar personalization with the automated, multi-dimensional assessment pipeline described in Section IV.

C. Bias, Litigation, and Regulation of AI Hiring Tools

A growing body of evidence documents fairness failures in commercial AI hiring tools. EPIC’s 2019 FTC complaint against HireVue argued that facial-analysis-based scoring was not scientifically validated and disproportionately disadvantaged candidates who did not match the system’s training distribution, including non-native English speakers and candidates with disabilities; HireVue subsequently dropped facial analysis in January 2021 [7]. The EEOC’s first AI-discrimination settlement involved iTutorGroup, where automated screening software allegedly rejected applicants on the basis of age [9]. In 2025, the ACLU of Colorado filed a complaint against Intuit and HireVue alleging that an AI-scored video interview relying on automated speech recognition disadvantaged a deaf, Indigenous candidate whose accommodation request was denied [8], [9]. These cases motivate treating disclosure, human oversight, and accommodation support as first-class design requirements rather than after-the-fact policy add-ons, which is reflected in the platform’s design (Section IV) and discussed further in Section VII.

On the regulatory side, Illinois has required employers to disclose AI use, explain how it works, and obtain candidate consent for AI-analyzed video interviews since 2020 [11]. NYC Local Law 144, in force since July 2023, requires an independent annual bias audit of any automated employment decision tool, publication of selection-rate and scoring-rate impact ratios by race, ethnicity, and sex, and at least ten business days of advance notice to candidates [10]. The EU AI Act designates employment-related AI systems, including CV-screening and interview-scoring software, as high-risk, imposing conformity assessment, transparency, and human-oversight duties on providers and deployers [11]. Any production deployment of the platform described here would need to be designed against these obligations from the outset.

III. SYSTEM ARCHITECTURE

The platform follows a client-server architecture with a clear separation of concerns, illustrated in Fig. 1. The frontend is a React single-page application (SPA) optimized for modern browsers. The backend consists of serverless functions and a Node.js server that interface with an external LLM provider and a document-oriented database for persistence.

A. Frontend Layer

The frontend is built with React 18 and Vite, styled with TailwindCSS, and organized into two portals:

- **Interviewer Portal:** dashboard, avatar training, session management, question creation, candidate tracking with AI-derived scores and rankings, and configuration settings.
- **Candidate Portal:** registration via unique links, the AI-conducted interview experience, live transcription, and progress tracking.

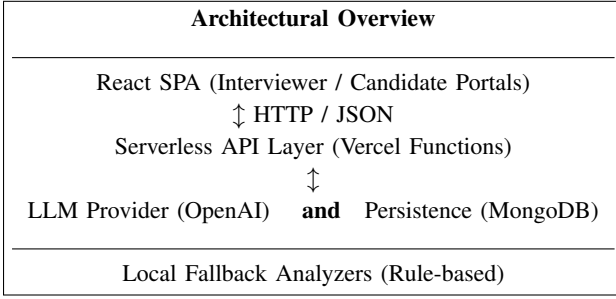


Fig. 1. High-level system architecture. When the LLM provider is unreachable or rate-limited, requests are transparently served by the local fallback analyzers instead of failing.

B. Backend Layer

The backend is implemented as serverless functions deployed on Vercel, supplemented by a Node.js server for local development. It exposes endpoints for session management, candidate registration, question generation, resume analysis, and response analysis, and mediates all LLM calls so that API credentials remain server-side and are never exposed to the browser.

C. Data Persistence

Session and candidate data can be persisted locally in the browser via the Web Storage API, or in MongoDB for cross-device consistency. This lets the platform run in offline or single-device demo scenarios while remaining deployable at production scale.

IV. METHODOLOGY

A. Personalized AI Avatar

HR professionals upload facial images and, optionally, a short demo video. During interviews, the avatar renders the interviewer’s image or video alongside TTS narration of each question, with rate and pitch tuned toward natural-sounding speech. Speech timing is synchronized with lightweight visual cues (a scaling animation and an active-speaking indicator). Candidates can replay questions, mute audio, and toggle subtitles, which is also relevant to the captioning-accommodation concerns raised in Section VII.

B. Speech and Media Processing

The platform uses the Web Speech API for synthesis and recognition. Candidate answers are captured with the MediaRecorder API as WebM audio/video, with speech recognition producing a live transcript for downstream analysis. The pipeline supports camera/microphone toggling, pause/resume, and automatic duration limits per question.

C. Question Generation

Question generation operates in two modes. In AI-assisted mode, a structured prompt combining the job description and optional resume text is sent to the LLM, which returns role-specific questions with expected keywords, time limits, and

per-question weights. In deterministic mode, a local rule-based generator extracts skills and project signals from the job description and resume to construct comparable questions without external connectivity. This dual-mode design guarantees the system remains usable when the LLM provider is unavailable.

D. Multi-Dimensional Response Scoring

Each answer is evaluated along five dimensions:

- **Relevance** (R): how directly the answer addresses the question, computed from lexical overlap plus a response-length adjustment.
- **Accuracy** (A): technical or practical correctness for domain-specific questions, derived from keyword matching combined with relevance.
- **Confidence** (C): clarity and assertiveness of the response, inferred from linguistic confidence markers and length.
- **Keyword coverage** (K): the proportion of expected concepts present in the answer.
- **Skill presence** (S): detection of competencies such as communication, problem-solving, teamwork, leadership, technical depth, and adaptability, via curated keyword lexicons.

The overall per-question score is

$$\text{Overall} = \min(100, w_R R + w_A A + w_C C + w_K K) \cdot W, \quad (1)$$

with weights $w_R = 0.25$, $w_A = 0.30$, $w_C = 0.20$, $w_K = 0.25$, and W a per-question importance factor set at question-creation time. S is reported alongside the numeric score as a competency profile rather than folded into Eq. 1, since skill presence is better presented qualitatively than compressed into a single scalar. When the LLM is reachable, an instruction-based prompt with a structured JSON schema produces calibrated R , A , C , K values together with evidence-based strengths and improvement notes; when it is not, the platform transparently falls back to the deterministic scorer described above. Algorithm 1 summarizes the response-scoring procedure, including the fallback path.

E. Candidate Ranking

After all questions are answered, the platform aggregates per-question scores into summary statistics (mean relevance, accuracy, confidence, and overall score) and sorts candidates by aggregate score to produce a ranked shortlist. We treat this ranking as a decision-support signal for a human recruiter rather than an autonomous accept/reject decision; see Section VII.

F. Phone-Usage Detection

To support interview integrity, the platform implements a lightweight client-side heuristic: downsampled camera frames are analyzed via connected-component labeling to detect dark, high-contrast rectangular regions with aspect ratios consistent with a handheld phone. When such a region is detected during active recording, a warning is surfaced to the candidate and the event is logged through a callback for later review by a

Algorithm 1 Response Scoring with Graceful Degradation

Require: candidate answer a , question q with expected keywords κ , weight W

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1: scored  $\leftarrow$  False
2: if LLM service reachable then
3:    $(R, A, C, K, \text{notes}) \leftarrow \text{LLMANALYZE}(a, q, \kappa)$ 
4:   if response well-formed then
5:     scored  $\leftarrow$  True
6:   end if
7: end if
8: if not scored then
9:    $R \leftarrow \text{LEXICALOVERLAP}(a, q)$ 
10:   $K \leftarrow |\kappa \cap \text{terms}(a)|/|\kappa|$ 
11:   $A \leftarrow f(K, R)$ 
12:   $C \leftarrow \text{CONFIDENCEHEURISTIC}(a)$ 
13:  notes  $\leftarrow \text{RULEBASEDFEEDBACK}(a, \kappa)$ 
14: end if
15:  $S \leftarrow \text{SKILLLEXICONMATCH}(a)$ 
16: Overall  $\leftarrow \min(100, w_R R + w_A A + w_C C + w_K K) \cdot W$ 
17: return (Overall,  $R, A, C, K, S, \text{notes}$ )

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TABLE I
TECHNOLOGY STACK

Layer	Technology
UI Framework	React 18, Vite 5
Styling	TailwindCSS 3
Routing	React Router 6
Icons / Charts	Lucide React, Recharts
Speech	Web Speech API (TTS/STT)
Media Capture	MediaRecorder API
AI/LLM	OpenAI (gpt-4o-mini)
Backend	Node.js / Vercel Serverless
Database	MongoDB

human. All processing runs in the browser, so raw camera frames are never transmitted to the server; only the detection event is recorded. We are explicit that this is a coarse heuristic, not a certified proctoring solution (Section VI).

V. IMPLEMENTATION

The system is implemented with a JavaScript/TypeScript toolchain, summarized in Table I.

A. Resilience and Graceful Degradation

Every AI-dependent feature is wrapped with a fallback (Algorithm 1). If the LLM endpoint errors out or times out, question generation reverts to the local generator and response analysis reverts to the deterministic scorer. This keeps the platform operational under network constraints, provider outages, or rate limiting, at the cost of the semantic richness the LLM path provides.

B. Privacy and Security

Video and audio are captured and processed client-side wherever possible; the backend receives only text transcripts and metadata needed for analysis, and the phone-detection

heuristic never uploads camera frames. LLM API credentials are held server-side only. These choices reduce, but do not by themselves guarantee, compliance with data-protection and AI-hiring regulation; Section VII discusses what additional controls a production deployment would need.

VI. EVALUATION

A. Functional Evaluation

We evaluated the platform against its functional requirements across both interviewer and candidate workflows. The avatar guided candidates through multi-question sessions end to end, and live transcription was usable in Chromium-based browsers, where the Web Speech API is best supported. Both question-generation modes (AI-assisted and deterministic) produced coherent, role-relevant questions, and the fallback path activated correctly under simulated LLM outages, confirming that the degradation strategy behaves as designed.

B. Scoring Behavior

The deterministic scorer produces reproducible outputs across repeated runs on the same input, which is useful as a stable baseline and as a fallback guarantee. The LLM-based scorer produces richer, evidence-linked feedback but is non-deterministic across runs by nature of the underlying model. We have not yet conducted a controlled study comparing either scorer’s outputs to human interviewer judgments; we report this as a limitation rather than a validated capability. Establishing predictive or concurrent validity against expert human ratings, ideally with inter-rater reliability statistics (e.g., Cohen’s κ or intraclass correlation) and a demographically stratified sample, is necessary future work before the scoring model can be treated as more than a decision-support aid.

C. Limitations

- **Speech recognition variability.** Recognition quality varies by browser, performing best in Chrome and less reliably in Safari and Firefox, which can systematically disadvantage candidates on those browsers or with accents or speech patterns underrepresented in the underlying recognition model’s training data [8].
- **Semantic ceiling of the fallback scorer.** The deterministic analyzer cannot capture semantic nuance and may under-score well-phrased answers that use synonyms rather than the exact expected keywords.
- **Phone-detection false positives/negatives.** The connected-component heuristic is a coarse approximation and will misfire under varied lighting or camera angles; it should be treated as a flag for human review, not an automated integrity determination.
- **No validated fairness or accuracy evaluation.** We have not yet run a bias audit of the kind required under NYC Local Law 144 [10], nor a study of scoring parity across protected groups. Given documented fairness failures in comparable commercial tools [7]–[9], this is a prerequisite for production use, not an optional extension.

- **Throughput constraints.** Candidate volume and concurrency are bounded by third-party LLM API rate limits and quotas.

VII. ETHICAL CONSIDERATIONS AND REGULATORY ALIGNMENT

Because this platform automates a legally sensitive decision process, we treat fairness and compliance as design requirements rather than a discussion appendix.

A. Disclosure and Consent

Following the approach required under the Illinois AI Video Interview Act [11], candidates should be informed before the interview that an AI avatar and automated scoring are in use, given a plain-language explanation of what is measured, and asked for consent; the platform’s session-creation flow is a natural place to enforce this rather than leaving it to individual deployers.

B. Accommodation Support

The captioning and subtitle toggles described in Section IV are directly relevant to the accommodation failure alleged in the Intuit/HireVue case [8], [9]: a deaf candidate’s request for human-generated captioning was reportedly denied. A production deployment should treat accommodation requests (captioning, extended time, alternative response modalities) as a first-class, auditable workflow rather than a manual exception process.

C. Bias Auditing

NYC Local Law 144 requires an independent annual bias audit computing selection-rate and scoring-rate impact ratios across race, ethnicity, and sex, benchmarked against the EEOC’s four-fifths rule, with results published [10]. The platform’s structured scoring output (Section IV) is well suited to this kind of audit because per-dimension scores are logged individually, but the audit itself has not yet been performed and must precede any production use in a covered jurisdiction.

D. Human Oversight

Consistent with the EU AI Act’s treatment of employment AI as high-risk [11], we position the platform’s ranking output as decision support for a human recruiter, not an autonomous accept/reject mechanism, and recommend that any deployment keep a human in the loop for final decisions, particularly for candidates flagged by the phone-detection heuristic.

VIII. CONCLUSION AND FUTURE WORK

We presented an AI-driven automated interview platform combining a personalized AI avatar, real-time speech processing, LLM-based question generation, and a five-dimension candidate scoring model with a deterministic fallback for resilience. The system demonstrates a feasible architecture for scalable, engaging asynchronous interviewing, but we are explicit that it is currently a systems and design contribution: we have not yet validated scoring accuracy against human judgment, nor audited it for demographic bias, and both are

necessary before any production hiring use, particularly given documented fairness and accommodation failures in comparable commercial tools [7]–[9] and the regulatory obligations now in force in jurisdictions such as New York City and Illinois and under the EU AI Act [10], [11].

Future work will proceed along three tracks. First, a human-subjects study comparing platform scores to blinded expert-interviewer ratings, with inter-rater reliability statistics and a demographically stratified candidate sample, to establish predictive validity. Second, a formal bias audit computing selection-rate and scoring-rate impact ratios across protected groups, following the NYC Local Law 144 methodology, prior to any production deployment. Third, technical improvements: speaker diarization and prosodic/emotion analysis to enrich soft-skill assessment (evaluated for fairness impact before adoption), a trained object-detection model to replace the current phone-detection heuristic, and multimodal fusion of video, audio, and transcript signals for a more holistic and better-calibrated candidate evaluation.

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